SECTION A: ATTEMPT ALL QUESTIONS. **(55 marks)**

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1) i) The main role of an operating system is to: **(2 marks)**

a) Play games

b) Manage hardware and software resources

c) Print documents

d) Store files only

ii) What is memory management in an operating system? **(2 marks)**

a) Managing the CPU's processing speed
b) Allocating and optimizing system memory for processes
c) Controlling user access to files
d) Managing data transmission in a network

2) Which statement correctly differentiates between GUI and CLI operating systems? **(2 marks)**

a) GUI uses graphical elements like icons and menus, while CLI uses text commands typed by the user.

b) GUI uses only text commands, while CLI uses graphics and icons.

c) GUI and CLI are exactly the same type of operating system.

d) GUI cannot run programs, while CLI can run programs.

3) What does NTFS stand for in terms of file systems? **(2 marks)**

4) Match the Operating Systems with their type/examples. **(4 marks)**

Column A (Operating Systems)	**Column B (Examples / Types)**
a) Microsoft Windows	i) Mobile Operating System
b) macOS	ii) Apple Desktop Operating System
c) Linux	iii) Open-source Operating System
d) Android	iv) Microsoft Desktop Operating System

5) State four functions of an operating system **(4 marks)**

6) Which statement correctly describes file management? **(2 marks)**

a) Controls CPU speed

b) Organizes and manages files

c) Controls hardware devices

d) Protects passwords only

7) Match the common backup devices with their descriptions: **(10 marks)**

Backup devices	**Descriptions**
a) Tape Drive	i) A portable storage device commonly used for quick data transfer.
b) Optical Discs	ii) A traditional method for long-term data storage.
c) USB Flash Drives	iii) A solid-state device used for fast read/write operations.
d) SD Card	iv) A removable storage medium often used in cameras and smartphones.
e) HDD/SSD	v) A disc-based storage option used for archiving data.

8) Which of the following best describes how processor management improves computer performance? **(2 marks)**

a) It controls how the CPU handles tasks by allocating processing time efficiently to different programs.

b) It increases the size of the monitor screen for better viewing.

c) It permanently stores all computer files in memory.

d) It connects the computer to the internet automatically

9) Which component is the brain of the computer? **(2 marks)**

a) RAM

b) Keyboard

c) Monitor

d) Processors

10) Explain the main purpose of antivirus software. **(4 marks)**

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11) Security management in an OS is used to: **(2 marks)**

a) Increase RAM
b) Delete files
c) Protect data from unauthorized access
d) Format disks

12) Minimum RAM for 64-bit Windows 10 is: **(2 marks)**

a) 1GB
b) 2GB
c) 4GB
d) 8GB

13) Choose the correct option for each statement below. **(2 marks)**

Job accounting data can be used for:
a) System planning and optimization
b) Playing games
c) Increasing screen resolution
d) Creating passwords automatically

14) Write True or False for each statement. **(4 marks)**

a) The user interface in an operating system act as a bridge between the user and the computer hardware……..

b) The user interface is only used for deleting files from the computer….

c) The user interface allows users to interact with the system and run programs…….

d) The user interface helps make the system easier and more efficient to use.

15) A student wants to avoid losing important files. What should they do? **(2 marks)**

a) Delete files
b) Backup data
c) Rename files
d) Compress files

16) What protects user accounts in a computer system? **(2 marks)**

a) Password
b) Speaker
c) Monitor
d) Scanner

17) Explain why security is an important feature of an operating system. **(5 marks)**

SECTION B: ATTEMPT ANY THREE (3) QUESTIONS (30 marks)

18) Fill in the blank space with the correct word from the following list: **(10 marks)**

(Operating System, RAM, Processor, File Management, Security, GUI, CLI,

Hardware, Software, Memory, Task Scheduler, Event Viewer)

a) A ..... is system software that manages computer resources.
b) ..... is used to store data temporarily while a computer is running.
c) The .....is the brain of the computer that processes instructions.
d) ..... refers to the physical parts of a computer system.
e) ..... refers to programs and instructions used in a computer.
f) ..... is used to organize and manage files and folders.
g) ..... protects a computer system from unauthorized access.
h) ..... stores data and instructions for processing.
i) A ..... interface uses icons, windows, and menus.
j) A ..... interface uses text-based commands.

19) Fill-in-the-Blank:

a) The process of retrieving and returning data from a backup to its original location or a new device is called..... **(2 marks)**

b) Match the backup type with its description **(8 marks)**

Backup Type	Description
a) Full Backup	i) Saves changes since last backup
b) Incremental Backup	ii) Saves everything every time
c) Differential Backup	iii) Creates an exact copy without history
d) Mirror Backup	iv) Saves changes since last full backup

20) Match items in **Column A** with the correct answers in **Column B**.

(10 marks)

Column A	**Column B**
a) Change desktop wallpaper	i) Display settings
b) View system information	ii) Language settings
c) Adjust screen resolution	iii) Control Panel
d) Set system password	iv) Personalization
e) Install a new program	v) Task Manager
f) Remove unwanted files	vi) Time settings
g) Manage startup programs	vii) Apps & features
h) Organize files into folders	viii) System properties
i) Set date and time	ix) Disk cleanup
j) Change keyboard language	x) File management

21) Fill in the blank space with the correct word from the following list:

(antivirus, firewall, password, backup, malware, security, protection, encryption, update, authentication) **(10 marks)**

a) A strong ..... helps prevent unauthorized access to a computer system.

b) ..... software is used to detect and remove viruses from a computer.

c) A..... is used to block unauthorized access to a network.

d) Regular ..... of data helps prevent loss of important files.

e) Computer ..... is important for keeping data safe.

f) ..... is harmful software that damages computer systems.

g) ..... helps to secure and protect computer information.

h) ..... is used to convert data into a secure form.

i) Software ..... helps keep the system safe from new threats.

j) ..... is the process of verifying a user's identity before access.

22) Briefly explain any five (5) types of operating systems from the list below: **(10 marks)**

a) Batch Operating System.
b) Distributed Operating System.
c) Real-Time Operating System (Hard and Soft).
d) Network Operating System.
e) Time-Sharing Operating System.
f) Mobile Operating System.
g) Embedded Operating System.

SECTION C: ATTEMPT ONLY ONE (1) QUESTION **(15 marks)**

23) Match Feature of operating system in column A with its correct purpose in column B **(15 marks)**

Column A (Feature)	**Column B (Purpose)**
1. Process Management	Protects system and data
2. Memory Management	Manages hardware communication
3. File System Management	Handles input/output devices
4. Device Management	Allocates/deallocates memory
5. I/O Management	Organizes and secures files
6. Security Management	Controls execution of programs
7. User Interface	Connects systems together
8. Multitasking	Distributes system resources
9. Networking	Enables user interaction
10. Error Handling	Manages simultaneous processes
11. Resource Allocation	Runs multiple apps at once
12. Concurrency Control	Detects and fixes issues
13. Performance Monitoring	Restores data after failure
14. Portability	Runs across hardware platforms
15. Backup & Recovery	Tracks system efficiency

24) Explain the factors that must be considered when identifying customer requirements for an operating system. **(15 marks)**